

Orthogonal Space

$$v \in \mathbb{R}^n$$

space orthogonal to $v \rightarrow v^\perp$

$$\Rightarrow \langle v, w \rangle = 0$$

↓
orthogonal vectors.

w will be a vector space

$$\cdot \langle v, w \rangle = 0$$

$$\cdot \langle v, cw \rangle = c \langle v, w \rangle = 0$$

$$\cdot \langle v, w + w' \rangle = \langle v, w \rangle + \langle v, w' \rangle = 0$$

$\{v_1, v_2, \dots, v_n\}$ orthogonal basis of $\mathbb{R}^n$

$v_1^\perp \Rightarrow \{v_2, \dots, v_n\} \rightarrow$ basis of $v_1^\perp$

Spectral Theorem

Every symmetric real matrix can be written as

$$A = Q \Lambda Q^T \rightarrow \text{orthogonal matrix}$$

diagonal as
eigenvalues.

$\Rightarrow$ Let's say q_1 is a real eigenvector of A .

$$W = q_1^\perp$$

Prove

$\Rightarrow \omega \in W$ then

$$A\omega \in W$$

$$\langle \omega, q_1 \rangle = \omega^T q_1 = 0$$

$$\langle A\omega, q_1 \rangle = (A\omega)^T q_1 = \omega^T A^T q_1$$

$$\Rightarrow \omega^T \underbrace{Aq_1}_{\rightarrow \lambda q_1} = \langle \omega, \lambda q_1 \rangle = \lambda \underbrace{\langle \omega, q_1 \rangle}_{=0} = 0$$

Exercises

$$\textcircled{1} A = Q \Lambda Q^T = \sum_i \lambda_i q_i q_i^T$$

$$\Rightarrow Q \rightarrow \begin{bmatrix} | & & | \\ q_1 & \dots & q_n \\ | & & | \end{bmatrix}$$

$$\Rightarrow Q \Lambda = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \dots & q_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & 0 \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

$$= \begin{bmatrix} | & | & & | \\ \lambda_1 q_1 & \lambda_2 q_2 & \dots & \lambda_n q_n \\ | & | & & | \end{bmatrix}$$

$$Q \Lambda Q^T = \begin{bmatrix} | & | & & | \\ \lambda_1 q_1 & \lambda_2 q_2 & \dots & \lambda_n q_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \text{---} q_1 \text{---} \\ \text{---} q_2 \text{---} \\ \vdots \\ \text{---} q_n \text{---} \end{bmatrix}$$

$$= \underline{\underline{\sum \lambda_i q_i q_i^T}} \quad \text{as } q_i q_j^T = 0 \text{ for } i \neq j$$

$$\begin{aligned}
 \textcircled{2} \quad \det(A) &= \det(Q) \det(\Lambda) \det(Q^T) \\
 &= \underbrace{\det(QQ^T)}_1 \det(\underbrace{\Lambda}_{\text{diagonal}}) \\
 &= d_1 \cdot d_2 \cdot \dots \cdot \lambda_n.
 \end{aligned}$$

Positive semi-definite matrices

- positive definite if
 $\nearrow \text{symmetric}$
 $x^T A x > 0 \quad \forall x \neq 0$

- positive semidefinite if
 $x^T A x \geq 0 \quad \forall x$

⇒ These are equivalent

① A is positive definite.

② All $\lambda > 0$

③ $\exists B$ such that $A = B^T B$
↓
invertible.

⇒ 2// $A = Q \Lambda Q^T$

$$\Rightarrow x^T A x > 0$$

$$\Rightarrow x^T Q \Lambda Q^T x > 0$$

$$\underbrace{(Q^T x)^T}_{y} \Lambda \underbrace{(Q^T x)}_y$$

$$\boxed{y^T \Lambda y > 0}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\Rightarrow [y_1 \ y_2 \ \dots \ y_n] \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\Rightarrow [\lambda_1 y_1 \ \lambda_2 y_2 \ \dots \ \lambda_n y_n] \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\sum_{i=1}^n \lambda_i y_i^2 > 0$$

$\Rightarrow$ Now we can choose x such that $y = \begin{bmatrix} y_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ and so on & so forth.

Then for every i we can individually say that

$$\lambda_i y_i^2 > 0$$

$$\boxed{\lambda_i > 0}$$

$$\underline{\underline{3}} \quad \text{let} \quad B = \Lambda^{\frac{1}{2}} Q^T$$

$$B^T = Q (\Lambda^{\frac{1}{2}})^T$$

$\hookrightarrow \Lambda^{\frac{1}{2}}$

$$\begin{aligned} B^T B &= Q \Lambda^{\frac{1}{2}} \cdot \Lambda^{\frac{1}{2}} \cdot Q^T \\ &= Q \Lambda Q^T = \underline{\underline{A}} \end{aligned}$$

$$\begin{aligned} \Rightarrow \det(B^T B) &= \det(B^T) \cdot \det(B) \\ &= \det(B) \cdot \det(B) \end{aligned}$$

$$\begin{aligned} \Rightarrow (\det(B))^2 &= \det(Q) \cdot \det(\Lambda) \det(Q^T) \\ &= \lambda_1 \cdot \lambda_2 \cdots \lambda_n \end{aligned}$$

$$\det(B) = \sqrt{\lambda_1 \cdot \lambda_2 \cdots \lambda_n} \quad \underline{\underline{\neq 0}}$$

Hence B is invertible.

Exercises

① λ_i of triangular matrix are its diagonal entries.

$$\Rightarrow \det(A - \lambda I) = 0 \quad \text{let } d_i \rightarrow i^{\text{th}} \text{ diagonal entry.}$$

$$\Rightarrow (d_1 - \lambda) \cdot (d_2 - \lambda) \cdots (d_n - \lambda) = 0$$

The roots of this polynomial are the eigenvalues.

$$\Rightarrow \lambda \in \{d_1, d_2, \dots, d_n\}$$

② over $\mathbb{C}$ $\overset{\text{symmetric}}{\text{tr}}(A) = \sum_i \lambda_i$ &
 $\det(A) = \prod_i \lambda_i$

$$\Rightarrow \det(A) = \prod_i \lambda_i \quad (\text{seen before})$$

$$\det(A - \lambda I) = \prod_i (\lambda_i - \lambda)$$

$$= (-\lambda)^n + \sum \lambda_i (-\lambda)^{n-1} + \dots$$

$$= (-\lambda)^n + \sum \lambda_i (-\lambda)^{n-1} \dots$$

$$\det(A - \lambda I) = \begin{vmatrix} a_{11} - \lambda & & \\ & a_{22} - \lambda & \\ & & \ddots \\ & & & a_{nn} - \lambda \end{vmatrix}$$

$$= \prod (a_{ii} - \lambda) + \text{terms in } \lambda^{\leq n-2}$$

⇒ Now compare $(-\lambda)^{n-1}$ coefficient as except $(a_{11} - \lambda)$ all other would have $\lambda^{\leq n-2}$

$$\sum \lambda_i = \sum a_{ii}$$

$$\boxed{\text{tr}(A) = \sum \lambda_i}$$

③ Prove Rayleigh - quotient bounds inequality for symmetric

$$\lambda_{\min} \leq \frac{x^T A x}{x^T x} \leq \lambda_{\max}$$

$$A = Q \Lambda Q^T$$

$$\Rightarrow x^T Q \Lambda Q^T x$$

$$y = Q^T x$$

$$\Rightarrow y^T \Lambda y$$

$$\Rightarrow y^T \Lambda y = \sum \lambda_i y_i^2$$

$$\begin{aligned} \Rightarrow x^T x &= x^T \underbrace{Q Q^T}_I x = y^T y \\ &= \sum y_i^2 \end{aligned}$$

$$\frac{x^T Q \Lambda Q^T x}{x^T x} = \frac{\sum \lambda_i y_i^2}{\sum y_i^2}$$

$\Rightarrow$ Convex combination rule (factored)

$$\lambda_1 \square + \lambda_2 \square \dots \lambda_n \square$$

$$\cdot \square \geq 0$$

$$\cdot \sum \square = 1$$

then

$$\lambda_{\min} \leq \sum \lambda_i \square_i \leq \lambda_{\max}$$

$$\Rightarrow \lambda_{\min} \sum \left(\frac{y_i^2}{\sum y_i^2} \right) \leq \text{our form} \leq \lambda_{\max}()$$

$\nearrow \text{our form} = 1$

Blocks of Matrices

$A, B, D, O \rightarrow$ blocks.

$$\begin{bmatrix} A & B \\ O & D \end{bmatrix} = \det(A) \cdot \det(D)$$

$\Rightarrow$ Show that geometric multiplicity of a λ is always less than or equal to algebraic multiplicity.

• geometric $(A - \lambda_0 I)x = 0$
 $\downarrow$
dimension of nullspace.

• algebraic $(\lambda - \lambda_0)^{\textcircled{n}} \dots$
 $\downarrow$
number of repeated roots

$$\Rightarrow \det(A - \lambda I) = (\lambda_0 - \lambda)^k \prod_i (\lambda_i - \lambda)$$

$$\Rightarrow \text{Null}(A - \lambda_0 I) \rightarrow \underline{\underline{G}}$$

$$G \rightarrow \text{basis of } \{v_1, v_2, \dots, v_g\}$$

$$\mathbb{R}^n \rightarrow \{v_1, v_2, \dots, v_g, \underbrace{v_{g+1}, \dots, v_n}_{G^\perp}\}$$

$$A = \begin{bmatrix} | & & | & & | \\ v_1 & \dots & v_g & \dots & v_n \\ | & & | & & | \end{bmatrix}$$

$\Rightarrow$ Read about proof

Orthogonal & Gram-Schmidt

$\{q_i\}$ is orthonormal when

$$\langle q_i, q_j \rangle = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

→ Show that $\{q_i\}$ are linearly independent

$$\sum c_i q_i = 0$$

$$\Rightarrow \sum c_i q_i \cdot q_k = 0 \quad k \in \{1, \dots, n\}$$

$$\Rightarrow \sum c_i \langle q_i, q_k \rangle = c_k \|q_k\|^2 = c_k \underset{=0}{=0}$$

$$c_k = 0 \quad \text{for any } k \in \{1, \dots, n\}$$

Hence $c_i = 0 \quad \forall i \in \{1, \dots, n\}$

Given a set of lin. independent vectors $\{a_1, a_2, \dots, a_n\}$, we can get a set of orthonormal vectors

$$\Rightarrow q_k = \frac{a_k - \sum_{j < k} \langle a_k, q_j \rangle q_j}{\|a_k - \sum_{j < k} \langle a_k, q_j \rangle q_j\|}$$